

■ 2.1.10 Advanced Topic: Levels in Expressions

Map applies functions at the first level in expressions; **MapAll** applies functions at all levels. Sometimes you may need finer control over where functions are applied.

Looking at the trees that correspond to expressions, one can define a general notion of “levels” in expressions. The level of a particular part of an expression is simply the distance down the tree at which that expression occurs. (The top of the tree is considered to be level 0.)

<code>Map[f, expr, n]</code>	apply <i>f</i> at all levels down to <i>n</i> in <i>expr</i>
<code>Map[f, expr, {n}]</code>	apply <i>f</i> just at level <i>n</i> in <i>expr</i>

Applying functions to various levels in expressions.

Here is a sample expression.

```
In[1]:= t = {{a b, c d^2}, {x^3 y^4}}
```

```
Out[1]= {{a b, c d^2}, {x^3 y^4}}
```

You can see the different levels in *t* from the tree form.

```
In[2]:= TreeForm[t]
```

```
Out[2]//TreeForm=
```

```
List[
  List[
    Times[a, b], Times[c, Power[d, 2]]
  ],
  List[
    Times[
      Power[x, 3], Power[y, 4]
    ]
  ]
]
```

This applies *f* at level 1, just as **Map[f, t]** would.

```
In[3]:= Map[f, t, 1]
```

```
Out[3]= {f[{a b, c d^2}], f[{x^3 y^4}]}
```

This applies *f* to level 2. Notice that *f* is nested exactly two deep.

```
In[4]:= Map[f, t, 2]
```

```
Out[4]= {f[{f[a b], f[c d^2]}], f[{f[x^3 y^4]}]}
```

This applies *f* *only* at level 2.

```
In[5]:= Map[f, t, {2}]
```

```
Out[5]= {{f[a b], f[c d^2]}, {f[x^3 y^4]}}
```

When you have got the hang of ordinary levels, you can try thinking about *negative levels*. Negative levels label parts of expressions starting at the bottom of

the tree. Level -1 contains all the “leaves” of the tree: objects like symbols and numbers.

This applies f to parts at the bottom level of the t tree.

```
In[6]:= Map[f, t, {-1}]
Out[6]= {{f[a] f[b], f[c] f[d]f[2]}, {f[x]f[3] f[y]f[4]}}
```

You can think of expressions as having a “depth”, equal to the maximum number of levels shown by **TreeForm**. In general, level $-n$ in an expression is defined to consist of all subexpressions whose depth is n .

Depth[expr] gives the “depth” of an expression.

```
In[7]:= Depth[x^2]
Out[7]= 2
```

This applies f to all parts of t that have “depth” 3. You may have to look quite carefully at this result to see exactly how it works.

```
In[8]:= Map[f, t, {-3}]
Out[8]= {{a b, f[c d2]}, {f[x3 y4]}}
```

Depth[expr]	the total number of levels in $expr$
Level[expr, lev]	a list of the parts of $expr$ at the levels specified by lev

Testing and extracting levels.

Level gives a list of all parts of an expression at particular levels. The parts of t at level -1 are all symbols or numbers.

```
In[9]:= Level[t, {-1}]
Out[9]= {a, b, c, d, 2, x, 3, y, 4}
```

This gives a list of the parts of t strictly at level -2.

```
In[10]:= Level[t, {-2}]
Out[10]= {a b, d2, x3, y4}
```

Here are the parts of t on levels 2 and below.

```
In[11]:= Level[t, {2, Infinity}]
Out[11]= {a, b, a b, c, d, 2, d2, c d2, x, 3, x3, y, 4,
          y4, x3 y4}
```

Here are the parts whose level specifications are between 2 and -2, inclusive.

```
In[12]:= Level[t, {2, -2}]
Out[12]= {a b, d2, c d2, x3, y4, x3 y4}
```

n	levels 1 through n
Infinity	all levels
$\{n\}$	level n only
$\{n_1, n_2\}$	levels n_1 through n_2

Level specifications.